

Accepted Manuscript

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PII: S2352-345X(17)30095-4
DOI: [10.1016/j.jcmgh.2017.05.007](https://doi.org/10.1016/j.jcmgh.2017.05.007)
Reference: JCMGH 248

To appear in: *Cellular and Molecular Gastroenterology and Hepatology*
Accepted Date: 26 May 2017

Please cite this article as: Gorelick FS, Lerch MM, Do animal models of acute pancreatitis reproduce human disease?, *Cellular and Molecular Gastroenterology and Hepatology* (2017), doi: 10.1016/j.jcmgh.2017.05.007.

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Grant support from: NIK R01 (DK54021), NIH P01 (DK098108), Veterans Administration Merit and Career development awards to FSG. Grant support from Deutsche Krebshilfe / Dr. Mildred-Scheel-Stiftung (109102), the Deutsche Forschungsgemeinschaft (DFG GRK1947, A3) the Federal Ministry of Education and Research (BMBF GANI-MED 03IS2061A and BMBF 0314107, 01ZZ9603, 01ZZ0103, 01ZZ0403, 03ZIK012), and the European Union (EU-FP-7: EPC-TM and EU-FP7-REGPOT-2010-1) to MML.

Conflicts of interests: Neither Dr. Gorelick nor Dr. Lerch have financial, personal, or intellectual conflicts relevant to the content of this publication.

Abstract

Acute pancreatitis is currently the most common cause of hospital admission among all non-malignant gastrointestinal diseases. To understand the pathophysiology of the disease and as a potential step toward developing targeted therapies, attempts to induce the disease experimentally began over 100 years ago. Recent decades have seen progress in developing new experimental pancreatitis models as well as elucidating many underlying cell biological and pathophysiological disease mechanisms. Some models have been developed to reflect specific causes of acute pancreatitis in humans. However, the paucity of data relating to the molecular mechanisms of human disease, the likelihood that multiple genetic and environmental factors affect the risk of disease development and its severity, and the limited information regarding the natural history of disease in humans makes it difficult to evaluate the value of disease models. Here we provide an overview of key models and discuss our views on their strengths for characterizing cell biological disease mechanisms or for identifying potential therapeutic targets. We also acknowledge their limitations.

Synopsis

Multiple animal models of clinical pancreatitis have been developed, however, limitations with respect to the pathobiology of human disease, makes it difficult to assess the validity of animal models. Though the growing availability of human pancreatic acinar cells are likely to provide information on early pancreatitis events that can be meaningfully compared to animal models, the relevance of animals to later events may be more difficult to obtain.

Introduction

Although models of acute pancreatitis have been used for several decades, the extent to which they recapitulate the events in humans remains unclear. The clinical course and limited pathologic material suggest that features of mild and severe pancreatitis are generally shared between these models and humans^{1,2}. Thus, mild and severe, and probably the new intermediate category of moderately severe disease, have similar courses and outcomes, though the time courses may differ with common rodent models usually progressing more rapidly than humans.

Some pancreatitis models are designed to examine issues related to the recognized causes of human disease and factors that might modulate disease severity. The major etiologies of acute clinical pancreatitis are gallstones, alcohol, and smoking. Procedure related, especially following endoscopic retrograde cholangiopancreatography (ERCP) acute pancreatitis (PEP) represents an important subgroup for which specific early interventions appear to reduce disease incidence and severity³. Renal failure⁴, diabetes, and especially obesity are substantial risk factors for disease development and severity. Though an infrequent cause of pancreatitis, exposure to potent cholinergic agonists as seen with scorpion bites⁵ and exposure to select insecticides^{6, 7}, can also cause disease and may be mimicked by preparations that use supraphysiologic concentrations of toxins or neurohumoral agents to induce disease⁸. Infections with Coxsackie virus can cause acute pancreatitis in children, though the incidence is unclear; this response has been reproduced in animal models⁹. Known genetic factors vary in their clinical impact. For example, mutations in cationic trypsinogen can alone cause pancreatitis, but such mutations make a very small contribution to the clinical load of this disease. Mutations in other molecules such as SPINK and CFTR appear to modify the risk of disease in the presence of other accepted factors, such as ethanol abuse. Though genetic models that reflect human variants have been generated, their value in reflecting clinical disease has been limited; we will not comment on these models further in this commentary.

Investigators striving to mimic these clinical etiologic factors in experimental models have focused on the underlying cellular mechanisms that are operative in these disease etiologies. In this respect, animal models have been remarkably successful in characterizing intracellular processes that precede tissue injury¹⁰⁻¹². Whether or not targeting these processes successfully in animal experiments predicts a similar beneficial effect in clinical trials remains another matter.

Animal models of clinical acute pancreatitis

Studies in experimental acute pancreatitis models are undertaken with several potential goals in mind. Most often is the wish to reproduce the mechanisms and processes underlying disease and/or examine therapeutic interventions, or use models to examine basic features of acute injury, inflammation, or tissue reconstitution. This means that the timing of an intervention needs to be relevant to the phase of disease. Models can also be used to examine relationship among organs during the course of acute injury. For example, how does pancreatic injury affect the lungs or the kidneys? Models can also explore factors that directly cause disease, sensitizers that affect the risk of developing disease or modulate its severity, or both. One lesson learned from both studies of models and human pancreatitis is that sequential and overlapping pathologic responses underlie this disease.

Models of acute pancreatitis study the disease at the cellular and whole animal level and in a range of species. Thus, isolated groups of pancreatic acinar cells have long been used to study basic cell physiology and more recently acute pancreatitis responses. Though they have the advantage of investigating responses in the absence of inflammation and changes in blood flow, their acinar cell phenotype disappears within hours of being placed in culture, making them useful for examining only early pancreatitis responses. Intact animal models of pancreatitis have been generated in many species including dogs, cats, guinea pigs, and even zebra fish^{13, 14}. Though many species have been used, rats and mice are the most commonly employed. This is because of their relatively low cost, the high reproducibility of rodent models, our potential to simulate the conditions we believe leads to human disease, and growing efficiency in manipulating gene structure and function. Here we focus on rodent models of acute pancreatitis and their potential relevance to human disease beginning with in vivo models and concluding with isolated cell preparations.

Although rodent models are most often used, their differences from human exocrine pancreas with regard to various factors, such as digestive enzyme content should be recognized. Experimental pancreatitis preparations vary in their difficulty, reproducibility, and apparent relevance to clinical disease¹⁴. Since the clinical relevance of most

models remains unclear, some studies have examined the efficacy of interventions using multiple models, which we view as wise^{15, 16}. Below we comment on some of the key experimental pancreatitis models.

Caerulein-induced pancreatitis: One of the oldest and most often used model of acute pancreatitis uses caerulein, a peptide orthologue of cholecystokinin, that when given in doses that are 10-100 times greater than a physiologic equivalent, causes mild acute pancreatitis in rats and a somewhat more severe variety in mice^{17, 18}. From 1 to 12 doses may be given hourly to induce disease. Though this also leads to lung injury, both the pancreatitis and lung injury are fully reversible within hours to a few days of the treatment. Mechanistically, this model is most similar to the pancreatitis caused by scorpion venom and cholinergic toxins that are thought to represent states of supraphysiologic neurohumoral stimulation. The model has been highly favored because of its strong reproducibility, simplicity, and the ease with which processes of intracellular protease activation can be studied^{19, 20}. It has also been extended to generate other forms of pancreatitis, such as the combination of caerulein with partial duct ligation which allows simultaneous investigations of acute reversible changes and progressive fibrosis within the same organ²¹. Another example, when caerulein is given in repeated doses over time, the model can be converted to one that generates either severe pancreatitis or one that has features of chronic disease^{22, 23}. One challenge with the caerulein model is that responses between mice and rats are different. For example, the time course of zymogen activation, the degree of inflammation, and pattern of cell death are distinct. Further, the ability of repeated doses of caerulein to cause fibrosis in a chronic pancreatitis model depends on the mouse strain²⁴. We believe that the caerulein model is likely relevant to initiator mechanisms in acute clinical pancreatitis, at least those that follow the earliest factors such as acinar cell receptor activation. It also seems probable that prophylactic or therapeutic interventions that fail to show benefit in this model, would not reduce injury in more severe models of acute pancreatitis. It is also the model best suited to identify or refute potential treatment targets in a cell biological context as it best preserves acinar cell physiology throughout the experimental disease course^{18, 25}.

Alcoholic pancreatitis: Though a number of models have used different avenues of delivery to generate alcoholic pancreatitis in rodents, the oral route given by diet or gastric infusion (and probably the former) probably best reflects human exposure. A striking feature of these models is that animals do not develop pancreatitis²⁶. Indeed, chronic alcohol feeding alone induces features of the endoplasmic reticulum stress response that protect against injury²⁷. However, such ethanol exposure sensitizes rodents to the development of acute pancreatitis²⁶. For example, rodents (rats and mice) chronically fed alcohol do not develop acute pancreatitis, but do so after co-treatment with physiological equivalents of caerulein²⁶. Whether the down-regulation of the cystic fibrosis transmembrane conductance regulator (CFTR) by ethanol ingestion, shown in both rodents and humans, is linked to this sensitization is unknown²⁸. Potentially of greater clinical relevance, chronic ethanol feeding sensitizes rats to lipopolysaccharide (LPS), a bacterial product that is a potent stimulant of innate immune responses that signals through the toll receptor, TLR4^{29, 30}. In this model of chronic alcohol ingestion in rats, LPS induces an acute pancreatitis response that persists and leads to chronic disease. However, continued exposure to alcohol is required to maintain the chronic pancreatitis response³¹. The ethanol-feeding LPS model may be one of the most relevant clinically. Elevated serum LPS levels are found in human with chronic alcoholism, thus potentially setting the stage for alcoholic pancreatitis to develop³². We conclude that the failure for ethanol ingestion alone to induce acute pancreatitis reflects the human condition, and an additional sensitizing factor, such as LPS, or a genetic predisposition is needed to cause disease in animal models.

Cigarette smoke induced disease: In the past decade, cigarette smoking has emerged as a risk factor for both acute and chronic pancreatitis^{33, 34, 35}. Epidemiologic studies suggest that its mechanism is different from alcohol. Animal models of pancreatic cigarette smoke toxicity are limited (reviewed in³⁶). Administration of liquid extracts of cigarette tobacco can cause mild pancreatic injury. Several have used nicotine or NNK, a metabolite of nicotine that appears to mediate some of the toxicity of cigarette smoke³⁷. NNK has also been found in pancreatic juice in concentrations up to

3 μ M. Given alone in vivo, nicotine and NNK causes mild pancreatic injury in short and long-term studies. NNK is a potent agonist of several cell surface receptors, including nicotinic receptors. In that context, using chemical inhibitors and mice with genetic deletions, several studies have found that pancreatitis effects of NNK (with acinar cells and in vivo) appear to be mediated by an α -7-nicotinic receptor³⁷. A very recent study suggested that one action of NNK in the pancreas was to reduce the activity of plasma-membrane and mitochondrial thiamine transporters^{38, 39}. This NNK affect was linked to reduced cellular mitochondrial function and ATP depletion. Though NNK is clinically important component of cigarette smoke, it is likely that other constituents contribute to disease. Thus, studies using rodents exposed to chronic cigarette smoke are needed to determine whether it alone can cause pancreatic injury, affect thiamine transporters and mitochondrial function or sensitize to other causes of acute pancreatitis, such as ethanol ingestion and/or LPS. Whether cigarette smoke alone can cause acute pancreatitis or act as a sensitizing factor to other insults, such as alcohol or LPS, are important but open questions.

Biliary pancreatitis: It has been firmly established that the presence of gallstones is a significant etiological factor for developing acute (but rarely chronic) pancreatitis, though the risk for gallstone carriers as a whole is minimal. It has also emerged that only gallstones small enough to migrate from the gallbladder into the duodenum can trigger disease onset⁴⁰. The underlying mechanism, however, has been much debated. In 1901 Opie postulated that impairment of the pancreatic outflow due to gallstone obstruction of the pancreatic duct causes pancreatitis⁴¹. This initial 'duct obstruction hypothesis' was somewhat forgotten when Opie published his second 'common channel' hypothesis during the same year⁴² which predicted that an impacted gallstone at the papilla could create a communication between the pancreatic and the bile duct (the so called 'common channel') through which bile would flow into the pancreatic duct and would cause pancreatitis. This latter hypothesis received much attention because, during surgery in the late course of the disease, bile stained pancreatic necrosis was often observed. Based on these observations, various models of biliary pancreatitis have been developed including one that uses low concentrations of glycodeoxycholic acid

infusion into the pancreatic duct that causes moderate to severe pancreatitis in the rat when combined with cerulein⁴³. This proposed disease mechanism remained popular in spite of a number of clinical inconsistencies such as that fact that in most humans the communication between the pancreatic duct and the common bile duct is much too short (< 6 mm) to permit biliary reflux into the pancreatic duct and that an impacted gallstone would most likely obstruct both the common bile duct and the pancreatic duct^{44, 45}. Even in the event of an existing anatomical communication, pancreatic secretory pressure would still exceed biliary pressure and pancreatic juice would flow into the bile duct rather than bile into the pancreatic duct^{46, 47}. In the context of this question, an animal model has greatly helped in elucidating the underlying disease mechanism. The American opossum (*Didelphis virginiana*) has a biliary-pancreatic anatomy that resembles that in humans, thus permitting the experimental conditions of bile duct obstruction, pancreatic duct obstruction, or simultaneous obstruction of both systems and reflux of bile into the pancreatic duct. In experiments using this opossum model it became clear that acinar cells are the initial site of injury during pancreatitis⁴⁸ and the impairment of flow from the pancreatic duct, rather than reflux of bile, triggers the disease onset⁴⁹. This insight was later confirmed by clinical studies involving human gallstone disease and pancreatitis^{50, 51, 52} and is now widely accepted as the mechanism of biliary pancreatitis⁵³. To address the underlying cell biological mechanisms how duct obstruction triggers pancreatitis investigators were restricted to studying intracellular organelle trafficking⁵⁴, and then resorted to rodent models to investigate relevant signaling events⁵⁵. This choice was made for the simple reason that opossum studies are inherently limited in scope due to a lack of information on protein structure, availability of antibodies, knowledge on genetic background and non-availability of inbred strains. The opossum model is thus an extreme example of how suitable an experimental model can be to address one narrow question regarding the pathogenesis of pancreatitis and our current limitations for studying other aspects of the disease. In this context, infusion of bile into the pancreatic duct under pressure does cause severe pancreatitis in animal models that is limited to the pancreatic head⁵⁶. Though this model does meet the requirement of causing severe experimental disease, it probably does not reproduce the conditions that initiate human biliary pancreatitis.

Obesity and pancreatitis: A subject in which close similarities between human and animal models may exist is related to the worsening effects of obesity, especially intra-pancreatic fat necrosis, on disease severity⁵⁷. Both diet-induced obesity and congenital forms of obesity are used to model the human state. Since the congenital models often also show changes in leptin metabolism, a factor that can influence inflammatory responses, their relevance to human disease is often unclear since obese humans do not have this defect. It seems likely that injurious factors released by both intra-pancreatic fat and peritoneal fat stores can worsen pancreatitis injury.

The importance of fat is reflected by pancreatic pathology: one of the earliest signs of histologic damage in pancreatitis is fat necrosis. Though obesity, especially the accumulation of fat in the abdomen, directly corresponds to disease severity, fat located within the pancreatic parenchyma appears to be especially deleterious. In an elegant study that used both human materials and rodent models, a correlation between the levels of pancreatic fat, extent of fat necrosis, and disease severity were shown⁵⁸. Based on studies done in cellular models and using information derived from necrotic human tissues, a model of damaging fatty-acid signaling between acinar and fat cells is proposed^{58, 59}. The length of free fatty acids as well as the degree of fatty acid saturation may also affect injury. Whether therapies aimed at reducing triglyceride hydrolysis by lipase to free fatty acids will be a useful therapy for clinical pancreatitis remains to be seen.

Nutrient-induced pancreatitis: CDE and arginine or other basic amino acids: Though the mechanism remains unclear, administering excess basic amino acids or derivatives of naturally occurring amino acids can cause severe acute and sometimes lethal pancreatitis⁶⁰. The mechanisms whereby these agents cause pancreatitis is unclear. Further, the CDE diet model shows selectivity, only causing severe disease in young female mice⁶¹. Among its other disadvantages, it has been difficult to avoid rapid severe responses in mice and the cost and care required to administer the CDE diet. Further, these extreme diets also cause damage to other organs such as the liver,

which may secondarily affect the pancreas. It is known that inherited disorders of branched-chain amino acid metabolism, as well as conditions of extreme or selective malnutrition, can greatly increase the risk of developing pancreatitis in humans⁶². It is therefore likely that diet-based models of pancreatitis have some relevance for human disease.

Post ERCP pancreatitis (PEP): Since the inception of this procedure, it has been recognized to cause acute pancreatitis in 2-15% of patients. The risk of developing this complication is related to features of the procedure and patient characteristics such as difficult cannulations, multiple injections and patients who are young, female, and have had previous pancreatitis. Rodent models involve securing a cannula in pancreatic duct and introducing contrast under pressure⁶³. Initial studies on this topic showed that standard formulations of contrast, all containing a synthetic iodinated molecule needed for radiographic visualization, caused pancreatitis in rats⁶³. A buffered solution of similar osmolality and pH, but lacking the iodinated compound, did not cause pancreatitis⁶³. Activation of TRPV1 neuronal receptors, which are pressure, heat, ethanol and pH sensitive and linked to neurogenic inflammation and pain, appeared to mediate this response^{64, 65}. More recent studies suggest that one effect of the iodinated compound relates to its ability to induce pathologic calcium signaling in both rodent and human tissues⁶⁶. Whether this effect is relevant to its activation of TRPV1 channels has not been explored. In these studies, pancreatitis responses were mild and the severe responses observed in a subset of human subjects were not observed. Though this preparation will likely provide useful mechanistic insights, its inability to cause severe PEP is a limitation. To improve the model, one option would be to determine whether a sensitizer factor such as LPS could shift these models to a more severe phenotype. Interestingly, though data suggesting that the non-steroidal anti-inflammatory drugs (NSAIDs) diclofenac and indomethacin reduce the incidence of PEP in humans, their effectiveness has not been confirmed in these animal models⁶⁷. However, since rectal administration of these NSAIDs may be needed for effectiveness in humans, it may be difficult to parallel such studies in animal models. The lack of a severe pancreatitis responses in current models may also limit the utility of these models.

Drug-induced pancreatitis: Another perceived shortcoming is the difficulty of studying drug-induced pancreatitis in animal models to gain insight into the underlying mechanisms of onset. This was true for the early phase of the AIDS epidemic when drug-induced pancreatitis was common and animal studies were found to be unhelpful⁶⁸. Similarly, the latest wave of anxiety about incretin-based antidiabetic agents has not been put to rest though the evidence for pancreatitis responses in models has been almost entirely negative^{69, 70}. Especially when idiosyncratic drug reactions are concerned, differences between rodent and humans in terms of genetic background, inherent metabolism and possibly microbiota may account for the limitations of animal models to identify underlying disease processes or characterize susceptibility factors.

General mechanisms, values and limitations of models and study design:

We feel confident that the general pathobiologic responses characterizing acute pancreatitis, including the early activation of digestive enzymes, release of inflammatory mediators, enhanced paracellular permeability in vascular and epithelial tissues, inflammation, pancreatic cell death by apoptosis and necroptosis, and lung damage are shared in rodents and humans. However, beyond superficial and select measurements, we often lack the data needed to state how similar the responses are – in part this is due to the cost of doing careful measurements of such markers over time, and in the case of human material, the lack of access to tissue with the exception of the rare samples obtained surgically. Further, most available tissue from humans is collected, if at all these days, well after the early stages of the disease and not during disease initiation. Additionally, the complexity of the disease in rodent models, the paucity of detailed data from humans, and the infrequency with which beneficial therapeutic interventions have come from studies in pancreatitis models contribute to our lack of critical knowledge as to their relevance in a preclinical setting. While significant progress has been made in identifying and characterizing the underlying pathophysiology and cell biology of pancreatitis using experimental models, to what extent the results in animal pancreatitis can predict the outcome of clinical trials that target the same disease mechanism remains an unanswered question. For example,

we anticipate that disordering of cytosolic calcium signaling will be highly conserved as an early pathologic response in the pancreatic acinar cell, but the upstream pathways that induce this response could differ between model animals and humans⁷¹. It is also likely that for an individual patient, multiple factors contribute to the risk of developing acute pancreatitis and/or modulate its severity. For example, epidemiologic studies show that the majority of alcohol abusers, smokers, and even individuals with common bile duct stones do not develop acute pancreatitis. Thus, differences in environmental factors and genetic makeup must influence which individuals with these risk factors will develop pancreatitis, but most probably remain unknown. Genetic studies tell us that a range of factors, such as those that are related to digestive enzyme metabolism in the acinar cell⁷²⁻⁷⁶, can impact our risk for pancreatitis. It is likely that future studies will describe other genetic factors that affect the risk of developing pancreatitis or modify its severity. However, it seems likely if not certain, that most of these are obscured in the animal models we use to explore human disease unless the specific genetic alterations are introduced in the respective animal strain, a task now more easily achieved by the CRISPR Cas-9 gene editing technology. Moreover, the genetic variations and environmental differences that affect human susceptibility are eliminated by the deep inbreeding and maintenance of strict environmental conditions characteristic of previous rodent models.

A shortcoming of experimental design is that most are short term and do not follow animals to late complications or examine tissue reconstitution, with the notable exception of some chronic models²¹. An example of the translational importance of the latter highlighted by the recent recognition that pancreatic islet beta-cell failure may be seen as a late complication in 20-30% of acute pancreatitis⁷⁷. Differences in the microbiota, metabolism, diet, and regulation of inflammation can be substantial between humans and rodents and affect disease mechanisms and phenotypes. Thus, a range of factors, including inherited genetic differences among species and study design can influence the value of animal models. One example is the significant role Toll-like receptor 4 has in regulating the severity in an animal model of pancreatitis, whereas in the human disease loss of function polymorphism appear not to have such an effect⁷⁸.

⁷⁹. Another is the emerging recognition from models that trypsin activation is not required for the development of acinar cell necrosis^{80, 81, 82}, whereas trypsin mutations (most probably associated with a gain of function) are clearly the dominant genetic risk factor for pancreatitis in humans.^{13, 83-85} It is likely that no model of pancreatitis will ever recapitulate all aspects of human disease, but individual aspects and mechanisms can be specifically addressed.

Another important issue is the timing of interventions with respect to disease onset. Prophylactic therapies are initiated before disease onset; therapeutic interventions are giving after disease onset. Post-ERCP pancreatitis occurs in 2-15% of individuals undergoing this procedure and represents one of the very few clinical scenarios where one can administer prophylactic therapies⁸⁶. Patients with other forms of pancreatitis often do not present for treatment until 24 hrs after the onset of symptoms. Thus, opportunities to block early factors in acute pancreatitis may have passed by the time a patient arrives at the emergency department. Challenges in designing rationale interventions in animal models include limited knowledge of time course of disease and major mechanisms in humans and rodents. For example, the appropriate time to use an agent that reduces neutrophil activation or modulates macrophage phenotype in an animal model as well as which parameters are the best measures of success remains unclear⁸⁷⁻⁸⁹.

It is widely held that injury to the pancreatic acinar cell is a key initiating step in acute pancreatitis. Over the past two decades, treatment of acinar cells with supraphysiologic concentrations of the cholecystokinin orthologue, cerulein, or cholinergic agonists such as carbamoyl choline, and bile salts have been used to induce injury in acinar cells⁹⁰⁻⁹². Pancreatitis responses in these cells include changes in calcium signaling with a loss of physiologic oscillations, a high spike in cytosolic calcium followed by a slow return to basal levels^{71, 93-95}. Potential interventions that affect acinar cell calcium signaling are discussed below. Changes in other signaling pathways, such as select protein kinase C isoforms, and activation of NFkB are also early acinar cell pancreatitis responses⁹⁶⁻⁹⁸. In rodent acinar cells, these are soon followed by intracellular activation of digestive

zymogens such as trypsinogen⁹⁹, stimulation of and elaboration of inflammatory mediators, membrane and cytoskeletal disruption¹⁰⁰⁻¹⁰², decreased mitochondrial function and oxidative stress^{93, 103, 104}, decreased autophagic flux^{105, 106}, reduced secretion^{92, 107}, and activation of cell-death pathways. Recent access to freshly isolated human tissue from surgical procedures, as byproducts of preparing pancreatic islets for transplantation, or archival fixed material, has shown that at least some of the rodent acinar cell pancreatitis responses are conserved. This includes disordered autophagy and calcium signaling^{16, 108}. Though much of the work in this area has only appeared in abstract form or is in progress, the preliminary studies suggest that key human pancreatitis responses in the pancreatic acinar cell will be conserved in rodent acinar cells.

Has anything been identified in animal models that is clearly applicable to human disease?

The most important reason to study animal models of acute pancreatitis is the hope that we can improve the natural history of disease in humans. This could take advantage of opportunities to give prophylactic therapy that blocks the onset of acute pancreatitis or therapeutic interventions that are given after disease onset to reduce its severity. However, there may only be a few examples where the many studies in animal models have positively informed clinical studies and patient outcomes or where a beneficial effect in experimental pancreatitis has translated into a beneficial effect in humans.

Although intravenous fluids are considered a corner-stone of acute pancreatitis therapy, relatively little has been done in animal models until recently to explore the effects of different types and amounts of intravenous buffers. Prompted by a preliminary study showing that lactated Ringers appeared to reduce inflammatory responses compared to normal saline in a cohort of patients with severe acute pancreatitis¹⁰⁹, several basic science studies in rodents have been undertaken. One showed that a low-pH environment, such as might occur in the pancreatic parenchyma early in the course of acute pancreatitis, could worsen disease¹¹⁰. This finding could be relevant to the low pH of normal saline (~5.3), especially when it is given in large volumes. Subsequent studies

have shown that lactate, acts on a G-protein receptor, GPR 81, to suppress critical innate immune responses in acute pancreatitis and lessens disease severity¹¹¹. We anticipate that emerging studies will examine the effects of administering other metabolic intermediates on acinar cell and inflammatory responses in acute pancreatitis and that clinical studies will optimize the rationale use of therapies that modulate a range of acute inflammatory responses in this disease. In this case (modulation on innate immune responses), the available studies suggest that animal models may reflect human responses and inform interventions¹¹². The effects of differing volumes of intravenous fluids also needs to be studied.

The central role of abnormal calcium signaling in the acinar cell, including a demonstration of such abnormalities in an animal model of biliary pancreatitis, have made this an attractive therapeutic target⁵⁵. Several recent approaches to reducing pathologic acinar cell calcium signaling hold promise for human therapy. First, increasing extracellular magnesium reduces pancreatitis responses in rodent acinar cells including pathologic calcium signaling and lessens the severity of in vivo pancreatitis in rodent models^{113, 114}; it is now being examined for its potential role in reducing PEP¹¹². Insulin has recently been shown to lower calcium levels caused by induction of pancreatitis responses in acinar cells and reduce other cellular pancreatitis injuries^{115, 116}. These actions appear to be linked to a plasma-membrane calcium pump. Whether such protective effects of insulin can be shown in in vivo preparations awaits study. More recently, drugs have been developed that reduce calcium entry into the acinar cell and also dramatically reduce pancreatitis responses in isolated human acini¹⁶. Since abnormal acinar cell calcium signaling is largely an early response, such interventions might be most useful to prevent PEP. However, the efficacy of these agents may extend to the inflammatory response and provide a rationale for their use in clinical pancreatitis.

One example of a medication that is generally accepted as beneficial in humans (PEP), but has relatively little data in rodent models to support its use in humans is indomethacin and related anti-inflammatory drugs. Almost 40 years ago, indomethacin

was found to be a potent inhibitor of Ca^{2+} -activated polymorphonuclear phospholipase A2 (PLA2)¹¹⁷. Soon after, PLA2 levels were found to be elevated in the serum of subjects with acute pancreatitis and the degree of the increase corresponded to severity¹¹⁸. Roles from PLA2 from inflammatory as well as acinar cells in the pathogenesis of acute pancreatitis was proposed. Over 100 research publications and reviews on the topic followed with emphasis on its role in renal and lung injury¹¹⁹. Differences were observed with the prophylactic administration of PLA2 inhibitors in severe pancreatitis models^{120, 121}. However, indomethacin showed little benefit in a rat model of acute pancreatitis compared to a selective Cox2 inhibitor. Indomethacin transiently, but prominently, reduced pancreatic blood flow in pigs and has also been found to reduce perfusion in other vascular beds¹²². To our knowledge, no tests of the relative effectiveness of different administration routes for indomethacin was studied prior to the initiation of clinical trials. None-the-less, several, but certainly not all, clinical studies of indomethacin have suggested a benefit in reducing the severity of post ERCP pancreatitis^{67, 123}. Although this appears to be one of the few successful drug interventions for this disease, it would be difficult to argue that the data from experimental pancreatitis models predicts this benefit. Thus, with relatively little experimental support, indeed with some negative studies, clinical investigators could show unpredicted treatment benefit in humans.

Conclusions: The issue of whether an acute pancreatitis model reflects human disease should be evaluated in the context of the issue being addressed. Thus, is it important to consider the goal of the studies, especially if it is to develop a therapeutic intervention that could improve the natural history of a disease. This can be done prophylactically, such as interventions to prevent post ERCP pancreatitis, and therapeutically, when the disease is already established and corresponds to the clinical conditions and practice. It can be argued that if critical pathways are shared among the common causes of acute pancreatitis and between humans and animal models, that agents that are effective in these settings would have value irrespective of the initiating factor. Thus, the precise pathways that initiate alcoholic and biliary pancreatitis are of considerable scientific interest, but if early and late mechanisms are shared by different etiologies of acute

pancreatitis, it may not be critical to precisely mimic the conditions that initiate disease in humans. Our information related to factors that initiate and perpetuate acute pancreatitis are still being investigated. Additional time and research is needed to produce the data needed to make highly informed comparisons between animal models and humans. Furthermore, a major challenge is the difficulty in performing clinical trials to confirm the value of potential therapies identified in animal models. Though acute pancreatitis is not an uncommon disease¹²⁴, its course is extremely variable and very difficult to predict, thus necessitating large study cohorts. Further, the factors that provide the prognostic information needed to efficiently conduct clinical trials are still evolving. In this context, model investigative units for carefully designed collaborative studies provide examples of how the field might best move forward. Animal models have greatly and successfully contributed to the elucidation of some of the fundamental disease mechanisms of pancreatitis and will continue to do so. Before they can consistently be translated into improved care and better outcomes for pancreatitis patients, the field needs time, better funding, and a greater focus on patient-relevant results.

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Acknowledgements:

Grant support from NIK R01 (DK54021), NIH P01 (DK098108), Veterans Administration Merit and Career development awards to FSG. Grant support from Deutsche Krebshilfe / Dr. Mildred-Scheel-Stiftung (109102), the Deutsche Forschungsgemeinschaft (DFG GRK1947, A3) the Federal Ministry of Education and Research (BMBF GANI-MED 03IS2061A and BMBF 0314107, 01ZZ9603, 01ZZ0103, 01ZZ0403, 03ZIK012), and the European Union (EU-FP-7: EPC-TM and EU-FP7-REGPOT-2010-1) to MML.

Figure 1: Examples of factors that make it difficult to assess the relevance of acute pancreatitis models:

1) Rodents

- a. Inbred strains and strain differences
- b. Variations in microbiome and diet
- c. Potential differences in key regulatory/ target proteins
- d. Differences in preparations/experimental conditions
- e. Incomplete information (time course, inflammatory responses, multi-organ effects)

2) Humans

- a. Inherent human genetic and epigenetic variation
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